

Magic Squares

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1 Introduction

An order n magic square is an n by n grid filled with integers such that the rows, columns, and the two largest diagonals all add up to one number: the magic constant. Shown below is an example of a order 3 magic square with 15 as its magic constant.

8	1	6
3	5	7
4	9	2

2 Types of Magic Squares

The magic square we described is known as the normal magic square. A normal n by n magic square is filled up with consecutive integers from 1 to n^2 . Based on the definition, it follows that the magic constant has to be $\frac{1+2+3+\dots+(n^2-1)+n^2}{n}$ or $\frac{n^2(n^2+1)}{2n}$, since the total sum of the numbers must be split up evenly between each row or column. However, there are many variations that open up new possibilities, such as the need for the numbers filling the grid to be consecutive, multiplying instead of adding, and more explained below.

2.1 Multiplicative Magic Squares

Multiplicative magic squares are ones that multiply instead of adding to a constant. If we took the left magic square shown below and raised 2 to the power of each number in the cells, we will get the multiplicative magic square on the right, where each row, column, and diagonal multiplies to 4096.

3	8	1
2	4	6
7	0	5

2^3	2^8	2^1
2^2	2^4	2^6
2^7	2^0	2^5

8	256	2
4	16	64
128	1	32

We can take any magic square and raise any number, x , to the power of each number inside it. In this case, we didn't necessarily need to use powers of 2, since multiplication across these rows,

columns, and diagonals involve adding up the exponents. We already know the exponents add up to a magic constant which means they multiply to the same power of x in the end.

2.2 Complementary Magic Square

When you already have a magic square, you can make a complementary magic square by subtracting each cell from $n^2 + 1$.

8	1	6
3	5	7
4	9	2

$10-8$	$10-1$	$10-6$
$10-3$	$10-5$	$10-7$
$10-4$	$10-9$	$10-2$

2	9	4
7	5	3
6	1	8

2.3 Associative Magic Square

In an associative magic squares, pairs of cells opposite the center sum to $n^2 + 1$. In other words, consider each cell as a point and the opposite as a reflection over the square in the center, in this case 13. In this square, 1 is across from 25, 15 is across from 11, 24 is across from 2, and so on.

1	15	24	8	17
23	7	16	5	14
20	4	13	22	6
12	21	10	19	3
9	18	2	11	25

2.4 Panmagic Square

In a panmagic square, all of the diagonals, when wrapped around, rows, and columns sum to the magic constant. For a diagonal to wrap around, it continues on a diagonal on the opposite side of the main diagonal. Let's take the square shown in the section above, which is also a panmagic square! In the diagram on the lower left, we start from blue with 1. Then we continue onto the diagonal with 4 squares starting with 14 and sum all the numbers up. Shown below are the wrapped diagonals with each color presenting a different diagonal. The magic constant in this case is 65 (sum from 1 to 25 is 325, $\frac{325}{5} = 65$)

1	15	24	8	17
23	7	16	5	14
20	4	13	22	6
12	21	10	19	3
9	18	2	11	25

1	15	24	8	17
23	7	16	5	14
20	4	13	22	6
12	21	10	19	3
9	18	2	11	25

3 Making Magic Squares

Using these magic squares from above, we can develop techniques for creating them.

A complementary magic square is a square is defined by taking an existing magic square and transforming it into another!

Additionally, we can take one magic square and transform it by rotating the square 90° around any corner. We can also draw a mid-line horizontally or vertically and reflect the numbers to the boxes opposite. These transforms preserve the value of numbers in each row, column, and diagonal and are still valid magic squares after. The following 4 magic squares were created from the leftmost square by preserving the numbers that are in each row or column. Notice that 16, 2, 3, and 13 all stay in a line.

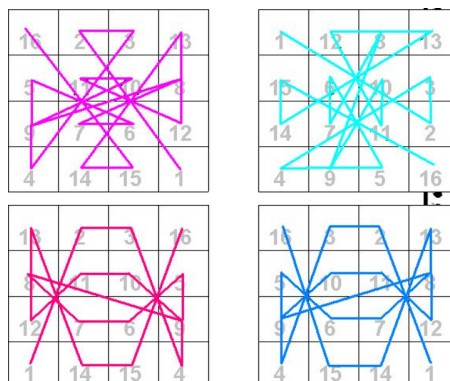
16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

1	12	8	13
15	6	10	3
14	7	11	2
4	9	5	16

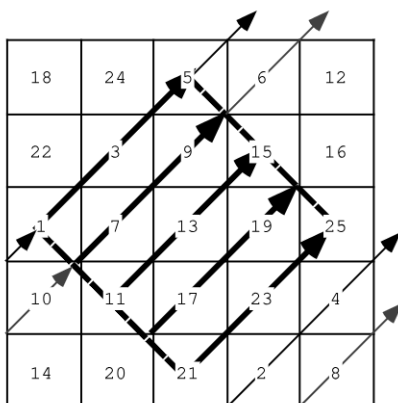
13	2	3	16
8	11	10	5
12	7	6	9
1	14	15	4

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

Note: When drawing a line through these magic squares from 1 to 16, they form a symmetrical pattern!

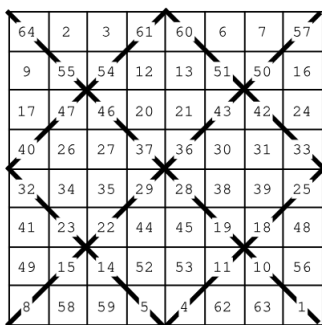


From the definition of panmagic squares, wrapped around diagonals all sum to a magic constant, it alludes to an approach that involves focus on the diagonals. One example of this called the Lozenge method, is shown below.



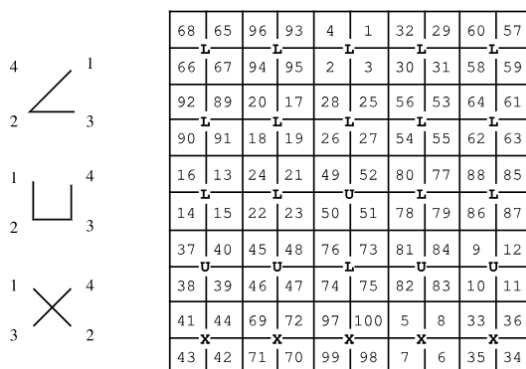
This starts with putting the numbers 1 through 5 in a wrap-around diagonal with odd numbers in a “diamond” in the middle, formed by the bolded lines, and even numbers outside. Then 6 through 10 are placed, 11 to 15, and so on.

A method using the idea of associative magic squares. Take an 8 by 8 grid, we fill in the grid with numbers 1 to 64 and draw these diagonals. Then we take the numbers along these diagonals that are opposite from each other, for example 1 and 64, and switch their positions. This involves the concept of associative magic squares with opposite numbers summing to $n^2 + 1$.



This diagram shows the end result after the swaps.

Some creative methods for producing order $4n + 2$, where m is an integer, normal magic squares such as the LUX method involve filling the magic squares with shapes. First, we start with an order $2n + 1$ magic square and $n + 1$ rows of Ls, 1 row of Us, and $n - 1$ rows of Xs. Then we would swap the adjacent L and U in the middle column which forms the letters shown below in the intersection of each group of four squares. Those letters refer to how each consecutive group of 4 integers will be arranged.



Let's try making a magic square with this method! For the sake of simplicity, we will consider a smaller case with an order 3 magic square which maps onto an order 6 normal magic square. We start with 0 and apply the L shape to get that the top middle group of 2 by 2. Then we move on to 1 to get the lower right group and so on.

7	0	5
2	4	6
3	8	1

L	L	L
L	U	L
U	X	U

Here is the final result you should get.

32	29	4	1	24	21
30	31	2	3	22	23
12	9	17	20	28	25
10	11	18	19	26	27
13	16	36	33	5	8
14	15	34	35	6	7

Taking from the idea of splitting large grids into smaller ones, the one below shows splitting a 6 by 6 grid into four quadrants of four groupings of consecutive numbers.

							165
8	1	6	26	19	24	84	
3	5	7	21	23	25	84	
4	9	2	22	27	20	84	
35	28	33	17	10	15	138	
30	32	34	12	14	16	138	
31	36	29	13	18	11	138	
111	111	111	111	111	111	57	

Only the columns line up right now, but with a few swaps between the cells, we can eventually get the rows to work. We can see that the sum of the first three rows and last three rows are constant which motivates us to swap numbers from rows 1-3 to rows 4-6 in the same column. As shown below, the yellow boxes swap with the orange boxes. Notice how the swaps occur 3 rows apart in the same column for the reasoning aforementioned.

						111
35	1	6	26	19	24	111
3	32	7	21	23	25	111
31	9	2	22	27	20	111
8	28	33	17	10	15	111
30	5	34	12	14	16	111
4	36	29	13	18	11	111
111	111	111	111	111	111	111

This strategy also works on a 10 by 10 grid, with the swaps looking recognizable as the pattern above, but with some extra steps involved.

										505
92	99	1	8	15	67	74	51	58	40	505
98	80	7	14	16	73	55	57	64	41	505
4	81	88	20	22	54	56	63	70	47	505
85	87	19	21	3	60	62	69	71	41	505
86	93	25	2	9	61	68	75	52	34	505
17	24	76	83	90	42	49	26	33	65	505
23	5	82	89	91	48	30	32	39	66	505
79	6	13	95	97	29	31	38	45	72	505
10	12	94	96	78	35	37	44	45	53	505
11	18	100	77	84	36	43	50	27	59	505
505	505	505	505	505	505	505	505	505	505	505

As we can see throughout the examples provided, magic squares of the same order don't have to look similar! Even similar ones are interesting to think about, for example, the rotations of the order 4 magic square shown before. Try playing around with creating magic squares and thinking about why each strategy works!

4 References

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